

NB16: Feature Engineering + Stacking (v2)

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Yaklasim

NB15 S1 kurgu (panel %50 + 625/625 MASTER cekirdek) + M3 missing uzerine: (1) FE ABLASYONU: no_fe (sadece M3) vs with_fe (Grantham/BLOSUM62/stopgain/log) -> FE'nin net katkisi izole olculur. (2) 6 base: scratch(lgbm,catboost,nn,dnn) + finetune(nn_ft,dnn_ft). (3) 2 meta: Logistic Regression + LightGBM. Threshold 3 mod, test %50/50 + bootstrap %80/20. Birincil: %80/20 F1.

Not: Finetune base OOF sadelestirmesi

nn_ft/dnn_ft OOF'ta MASTER-pretrain BIR KEZ yapilir (fold-bagimsiz, panel test'e degmez -> leakage minimal), finetune fold'a ozgudur. Tam-saf OOF degil; pragmatik.

1. FE Katkisi (panel-ort %80/20 F1)

Panel	no_fe	with_fe	delta
CFTR	0.5860	0.6547	+0.0687
KANSER	0.6665	0.6812	+0.0148
PAH	0.4587	0.4559	-0.0028

2. with_fe - %80/20 F1 (panel x model)

Model	CFTR	KANSER	PAH
catboost	0.8140	0.6894	0.4684
dnn	0.8520	0.6465	0.4468
dnn_ft	0.5450	0.7046	0.4490
lightgbm	0.5802	0.6666	0.4686
nn	0.5805	0.6378	0.3914
nn_ft	0.6610	0.6835	0.4377
stack_gbm	0.5227	0.7050	0.4703
stack_lr	0.6820	0.7164	0.5151

3. Train Sonuclari (with_fe, f1_8020) + Train-Test farki

Panel	Model	Tr-F1	Tr-P	Tr-R	Te-F1(50/50)	Tr-Te farki
CFTR	catboost	0.756	0.830	0.694	0.875	-0.119
CFTR	dnn	0.709	0.746	0.676	0.875	-0.166
CFTR	dnn_ft	0.822	0.826	0.819	0.795	+0.028
CFTR	lightgbm	0.724	0.816	0.651	0.757	-0.033
CFTR	nn	0.722	0.730	0.713	0.857	-0.136
CFTR	nn_ft	0.804	0.874	0.745	0.821	-0.016
CFTR	stack_gbm	0.839	0.974	0.737	0.686	+0.154
CFTR	stack_lr	0.799	0.890	0.725	0.750	+0.049
KANSER	catboost	0.778	0.840	0.724	0.896	-0.118
KANSER	dnn	0.692	0.799	0.610	0.809	-0.116
KANSER	dnn_ft	0.752	0.896	0.649	0.831	-0.078
KANSER	lightgbm	0.749	0.825	0.686	0.864	-0.115
KANSER	nn	0.735	0.769	0.704	0.866	-0.131
KANSER	nn_ft	0.837	0.894	0.787	0.867	-0.030
KANSER	stack_gbm	0.863	0.978	0.773	0.862	+0.002
KANSER	stack_lr	0.844	0.900	0.794	0.903	-0.060
PAH	catboost	0.763	0.846	0.696	0.883	-0.120
PAH	dnn	0.734	0.756	0.714	0.888	-0.154
PAH	dnn_ft	0.846	0.887	0.809	0.847	-0.001
PAH	lightgbm	0.771	0.845	0.709	0.887	-0.116
PAH	nn	0.749	0.760	0.738	0.885	-0.136
PAH	nn_ft	0.837	0.865	0.811	0.844	-0.007
PAH	stack_gbm	0.874	0.975	0.792	0.809	+0.065
PAH	stack_lr	0.842	0.903	0.788	0.872	-0.030

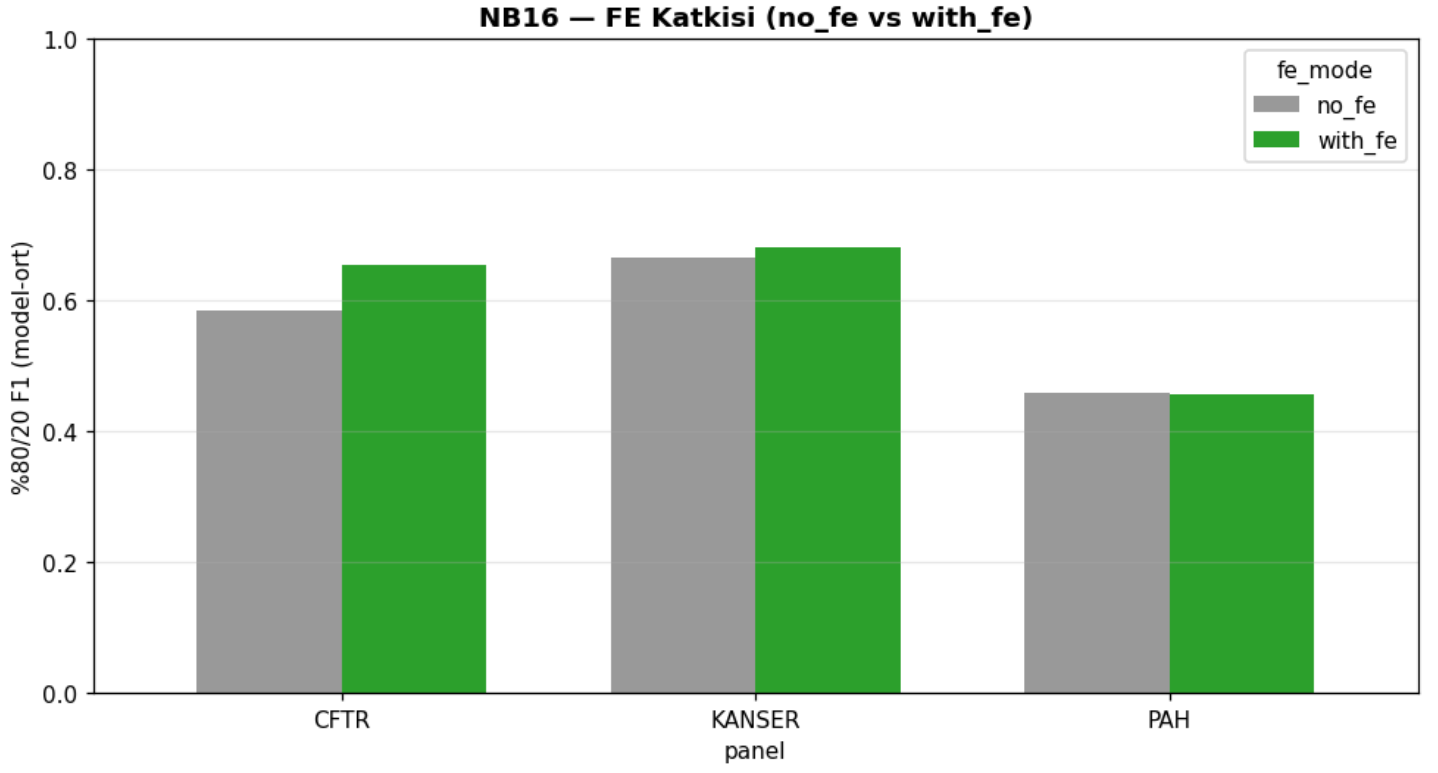
4. Stacking vs En İyi Base (with_fe, %80/20 F1)

Panel	En iyi stack	En iyi base	Kazanan
CFTR	stack_lr (0.6820)	dnn (0.8520)	base
KANSER	stack_lr (0.7164)	dnn_ft (0.7046)	stack
PAH	stack_lr (0.5151)	lightgbm (0.4686)	stack

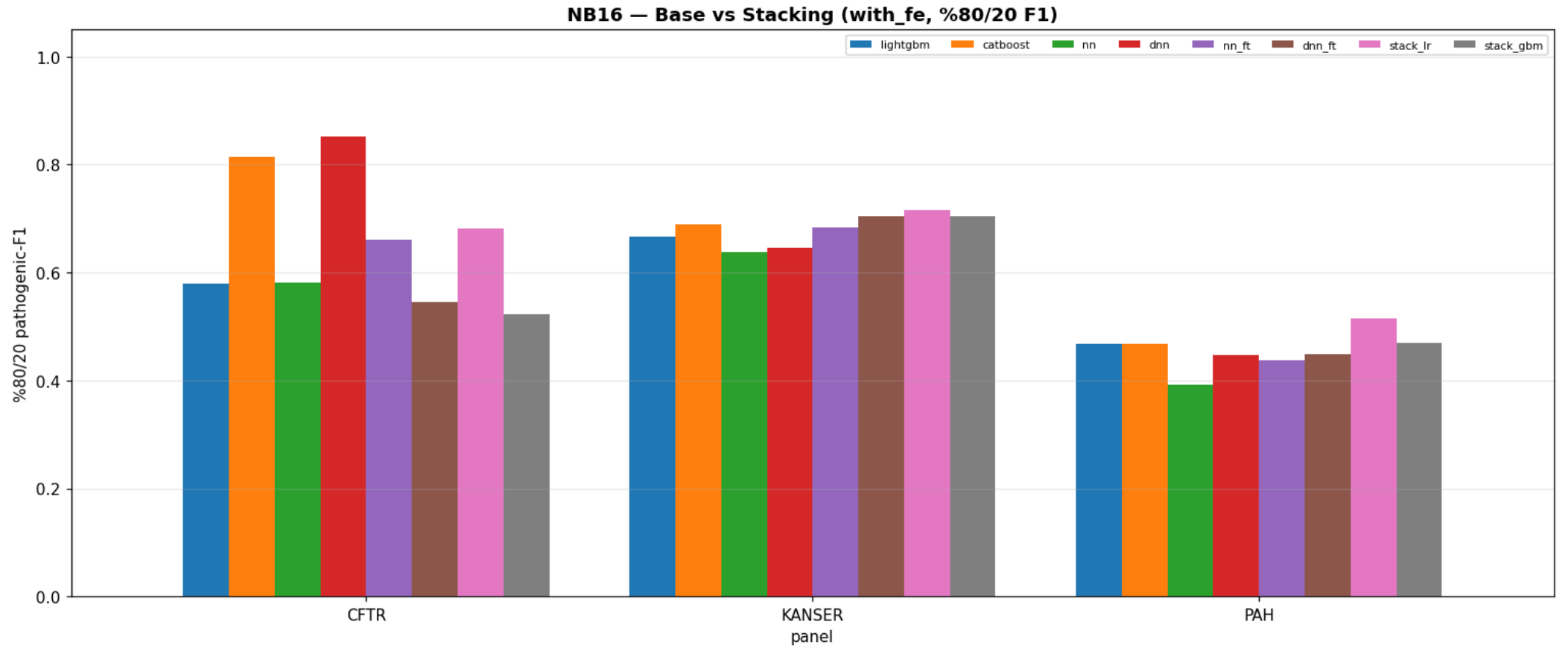
4b. Meta-learner (with_fe, panel-ort %80/20 F1)

Meta	Ort. %80/20 F1
stack_gbm	0.5660
stack_lr	0.6379

5. FE Ablasyonu (no_fe vs with_fe)



6. Base vs Stacking (with_fe)



7. En İyi Stacking Confusion Matrix

NB16 — En İyi Stacking Confusion Matrix (with_fe)

